

Load-Robust, Physics-Informed Machine Learning for Stator Inter-Turn Fault Diagnosis in Induction Motors: A Leakage-Safe Cascade for Detection, Severity, and Faulted-Phase Identification

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Abstract—Stator winding turn-to-turn (inter-turn) short-circuit faults are a leading cause of induction-motor failure, and early, load-robust diagnosis from non-invasive stator-current measurements remains an open problem. Most reported machine-learning (ML) studies achieve near-perfect in-distribution accuracy but never test generalization to unseen operating points, rarely release code, and seldom use leakage-safe evaluation. This paper presents a complete, openly released, physics-informed framework that performs hierarchical diagnosis from the three-phase stator current—fault-onset detection, healthy/faulty classification, ordinal severity estimation (percentage of shorted turns), and faulted-phase identification—and that is evaluated with a strict group-wise protocol and, above all, with *Leave-One-Load-Out* (LOLO) cross-validation. A 357-case PSCAD/EMTDC corpus ($f_s = 25$ kHz, $f_0 = 50$ Hz) is segmented into 18,033 labelled windows described by 72 physics-grounded features. LOLO reveals a clean structural result: detection (macro-F1 0.95) and faulted-phase identification (macro-F1 0.997) are intrinsically load-robust because they ride on load-invariant negative-sequence signatures, whereas absolute severity is load-dependent. We then (i) benchmark *eight* models on identical splits—including the tabular foundation model *TabPFN-2.5* and an *xLSTM* sequence network—establishing that, in-distribution, the task is saturated (all feature models within ± 0.01 macro-F1) so that the honest discriminator is cross-load behaviour; and (ii) introduce and ablate four load-robustness mechanisms: a sequence-component Stockwell tensor (cross-load severity within-one $0.75 \rightarrow 1.0$), a load-invariance-regularized feature selection, a FiLM load-conditioned multi-task network (cross-load severity $0.93 \rightarrow 0.98$), and a physics-anchored domain adaptation that recovers a simulation-trained detector on a public real-motor dataset ($0.48 \rightarrow 0.98$ macro-F1 with two labelled real samples). The deployed cascade reaches within-one-level accuracy of 0.95 in-distribution and 0.92 across unseen loads, with calibrated detection (expected calibration error 0.021), graceful noise degradation, and 100% fault-onset detection at 0% pre-fault false alarm. A within-hardware evaluation of all eight models on the real motor confirms detection transfers (macro-F1 up to 0.99) and severity is usable; per-window faulted-phase localization is the main gap (macro-F1 0.72 vs. 0.97 in simulation) but per-recording aggregation recovers it to 0.93–0.95—all reported transparently. Finally, leakage-safe Optuna tuning with a cross-load objective and physics-consistent, signature-preserving training-fold augmentation, evaluated under 10-fold grouped cross-validation, further raise cross-load severity (within-one $0.91 \rightarrow 0.94$; in-distribution $0.93 \rightarrow 0.98$; mean absolute error -20%) and detection (LOLO $0.96 \rightarrow 0.98$). A novel

block of biomedical anomaly features (signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG analysis) likewise lifts cross-load detection ($+0.03$) and severity ($+0.04$) where absolute physics magnitudes do not transfer. All code, splits, and artifacts are released.

Index Terms—Induction motor, inter-turn fault, stator winding, fault diagnosis, severity estimation, machine learning, negative-sequence current, leakage-safe evaluation, foundation model, transfer learning.

I. INTRODUCTION

A. Motivation

INDUCTION motors drive most industrial processes, and stator-winding insulation degradation is among their most common failure modes. An incipient inter-turn (turn-to-turn) short circuit (ITSC) shorts a small fraction of one phase's turns; the circulating fault current accelerates local heating and, if undetected, escalates to a catastrophic phase-to-ground or phase-to-phase fault [1], [2]. Surveys attribute roughly a third of IM failures to stator insulation, and the incipient-to-destructive interval can be short, so an on-line monitor must be both sensitive to small faults and trustworthy enough to trigger maintenance without excessive false alarms. Current-based monitoring (MCSA) is preferred over vibration or thermal sensing because the current is already measured at the drive, needs no extra instrumentation, and is robust to mounting. Three questions must be answered—*is* a fault present, *how severe*, and *which phase*—and the answers must hold up as the mechanical load changes.

B. Literature Review

Penman *et al.* [1] established the ITSC sideband frequencies; Cardoso *et al.* [2] introduced the Extended Park's Vector Approach (EPVA), where the $2f_0$ Park-modulus component scales with winding asymmetry. The negative-sequence current and negative-sequence impedance are widely used asymmetry descriptors [4], [5], with phasor compensation suppressing baseline supply unbalance. Mazzeletti *et al.* [6] use the positive-sequence fifth-harmonic during soft-starter start-up to distinguish ITSC from high-resistance connections; Singh and Shaik [7] combine Stockwell-transform features

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Code, dataset splits, and result artifacts are publicly available at <https://github.com/Regal-s/induction-motor-fault-diagnosis>.

with SVMs. Data-driven studies feed STFT symmetrical-component amplitudes to SVM/MLP classifiers [8], compare ANN/LSTM/physics-informed networks on a public dataset [9], or ensemble CNNs on spectrograms [10]. Recent architectures (TCN [38], attention [37], selective state-space [20], KAN [21], tabular foundation models) are mostly demonstrated on vibration or generic tabular data rather than three-phase stator current. The recent review [3] flags three systemic weaknesses: suspiciously perfect accuracies from leaky splits, scarce code/data releases, and little validation across operating conditions—the load dependence of the negative-sequence indicator is rarely confronted in a reproducible manner, and operating-condition shift is usually treated as a nuisance to be erased rather than a signal to be exploited.

C. Summary of Contributions

This paper treats ITSC diagnosis as a hierarchical problem (onset $\rightarrow$ detection $\rightarrow$ severity $\rightarrow$ faulted phase) and adopts *Leave-One-Load-Out* (LOLO) evaluation as its central methodological device. Concretely, the delivered system and its contributions are:

- A *leakage-safe evaluation framework*—grouping by operating point and using LOLO as the primary metric—together with a SHAP/Fisher analysis that explains *why* detection and faulted phase are load-invariant while absolute severity is load-dependent. We argue this should be standard practice for the field.
- A *four-stage cascade* on 72 physics-grounded features: onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification, with calibrated probabilities, conformal severity intervals, and an online onset trigger.
- A *controlled eight-model benchmark* on identical splits—logistic regression, SVM-RBF, k -NN, MLP, random forest, XGBoost, the tabular foundation model *TabPFN*-2.5, and an $xLSTM$ sequence network—showing the in-distribution task is saturated and that cross-load behaviour, not in-distribution accuracy, is the meaningful axis of comparison.
- A *novel biomedical anomaly-feature* block—signal-complexity, entropy, fractal-dimension, and heart-rate-variability-style cycle-to-cycle measures imported from ECG/EEG/HRV analysis, computed on the current residual and Park-vector modulus—that complements the physics features and improves cross-load detection (+0.03) and severity (+0.04) where absolute-magnitude features do not transfer.
- Four *load-robustness mechanisms*, each ablated under LOLO: a sequence-component Stockwell tensor (SCST) representation, a load-invariance-regularized feature selection (LIR-mRMR), a phase-coupled FiLM *load-conditioned* multi-task network (PCM-Net) that injects rather than erases the operating point, and physics-anchored domain adaptation (PADA) validated simulation-to-experiment, plus an honest negative result on physics-constrained generative augmentation.
- A *within-hardware evaluation* of all eight models on a real motor (leakage-safe, by recording and by repetition),

establishing that detection transfers (macro-F1 up to 0.99) and severity is usable, and transparently identifying faulted-phase localization as the main simulation-to-reality gap.

- A fully *reproducible* release of code, leakage-safe splits, feature tables, and per-model result artifacts.

A unifying theme emerges: the mechanical load is the crux of stator-fault diagnosis, and every proposed mechanism independently improves *cross-load* performance.

II. PROPOSED METHODOLOGY

A. Overview

The system ingests a short three-phase current window and produces, in sequence, a fault-onset estimate, a healthy/faulty decision, and—when faulty—an ordinal severity and the faulted phase. The four questions are nested, have different decision boundaries, and have different load sensitivities; we therefore use a hierarchical, gated architecture where the severity and phase stages are trained and operated only on data they are meant to explain. The thread running through every component—the symmetrical-component representation, the selection criterion, the model conditioning, and the transfer alignment—is load-robustness, which the Leave-One-Load-Out protocol of Section IV verifies. Fig. 1 renders the end-to-end flow.

Pipeline steps. For each 80 ms window $\mathbf{i}(t) = [i_a, i_b, i_c]^T$: (S1) acquire the three currents from the simulation or live ADC, indexed against the operating-point label and the verified region bounds (motor start, load change, fault, clearing; Fig. 2); (S2) slide $W=2000$ samples (4 cycles, 80 ms) at 75% overlap inside the fault interval, labelling the pre-fault region of every faulted record as healthy to obtain matched-load healthy data at every load; (S3) divide all three phases by a single per-recording RMS scale (removes load amplitude, preserves inter-phase imbalance); (S4) compute symmetrical components $|I_0|, |I_1|, |I_2|$, ratios $|I_2|/|I_1|, |I_0|/|I_1|$, and angles $\angle I_1, \angle I_2$ (with sin/cos for wrap-free use) via Clarke and Fortescue; (S5) the EPVA severity factor (amplitude of the $2f_0$ line of the Park-vector modulus over its DC), Park-ellipse descriptors, per-phase third-harmonic ratio, THD, RMS / skew / kurtosis and inter-phase differences, plus their $|I_1|$ -normalised *load-invariant* n_s variants; (S6) 24 biomedical anomaly features (HRV cycle-to-cycle distance, Poincaré SD1/SD2, Hjorth, permutation / spectral / sample / dispersion entropy, Higuchi / Katz / Petrosian fractal dimension) on the residual and Park modulus; (S7) group windows by operating point so no recording appears in both train and test, and apply physics-consistent augmentations to the training fold only (§II-F); (S8) Stage 0 onset trigger $\rho(k) = |I_2(k)|/|I_1(k)| > \bar{\rho}+6\sigma$ for three consecutive cycles, essentially free, gating the cascade in deployment; (S9) Stage 1 class-weighted XGBoost on the 96-feature bank with SHAP audit (ECE 0.021, §IV-I)—if healthy, the cascade terminates; if faulty, S10 and S11 fire in parallel; (S10) Stage 2 ordinal regression on $r \in \{0, \dots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}\%$ using the load-invariant subset with the *measured* load (load-aware severity); (S11) Stage 3 three-class phase head on the full feature bank, load-invariant by

Stator Inter-Turn Fault Diagnosis Pipeline (Track A, deployed)

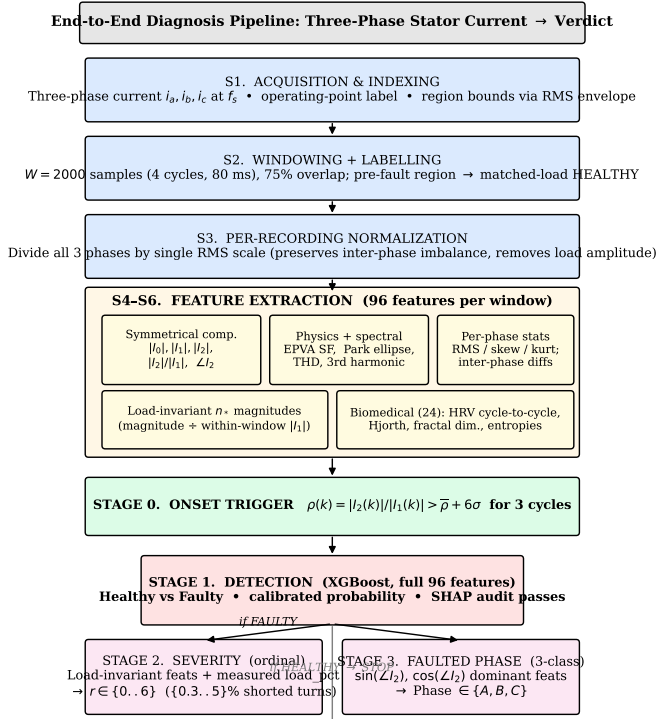


Fig. 1. Block diagram of the deployed diagnosis pipeline (steps S1–S12 of §II-A). Three-phase stator current enters at the top; per-recording normalization (S3) and the 96-feature extractor (S4–S6) feed Stage 0 (onset trigger on the per-cycle $|I_2|/|I_1|$ index) and the gated XGBoost cascade—Stage 1 detection → {Stage 2 severity (load-aware ordinal), Stage 3 faulted phase (3-class)}. Per-recording aggregation (S12) produces the final verdict.

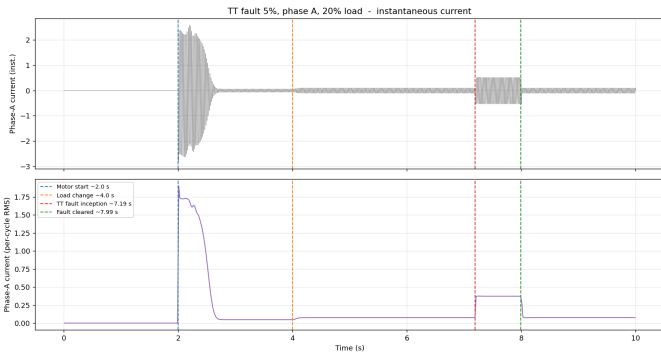


Fig. 2. Phase-A current (top) and per-cycle RMS envelope (bottom) of a faulted record, with motor start (2.0 s), load change (4.0 s), fault inception (7.2 s), and clearing (8.0 s). The fault interval supplies faulty windows; the pre-fault loaded region supplies matched-load healthy windows.

virtue of $\angle I_2$ (§IV-B); (S12) per-recording aggregation (majority vote for phase and state, median severity rank) absorbs per-window noise, recovering per-window phase macro-F1 from 0.72 to 0.93–0.95 on real hardware (§IV-J).

TABLE I

WINDOW-LENGTH TRADE-OFF AT $f_s = 25$ kHz. Δf IS THE SPECTRAL RESOLUTION OF (3). N_{FAULT} IS THE TYPICAL NUMBER OF NON-OVERLAPPING WINDOWS PER FAULTED RECORDING IN THE 0.8 s FAULT INTERVAL. “EPVA / HARMONICS USABLE” REQUIRES $\Delta f \ll 2f_0$.

Cycles	W	T_{win}	Δf	N_{fault}	EPVA / harm.
1	500	20 ms	50.0 Hz	40	no (bins merge)
4	2000	80 ms	12.5 Hz	10	yes (chosen)
10	5000	200 ms	5.0 Hz	4	yes, but redundant
20	10,000	400 ms	2.5 Hz	2	yes; too few examples

B. Signal Preprocessing and Symmetrical-Component Decomposition

Each recording is segmented into windows of $W = 2000$ samples (four 50 Hz cycles); a 75% overlap is used inside the short 0.8 s fault interval and a sparser stride elsewhere. Crucially, the *pre-fault* loaded region of every faulted record (genuinely healthy, since the fault is inserted only at 7.2 s) is labelled healthy, providing matched-load healthy data at the same loads as the faulty windows and preventing the detector from separating classes by load level. To remove load-induced amplitude while preserving the inter-phase imbalance that encodes the fault, all three phases of a window are divided by a single per-recording scale; magnitude features are additionally divided by the within-window positive-sequence fundamental $|I_1|$ to obtain load-invariant variants. The Clarke transform underlying the EPVA and the Park ellipse is

$$i_\alpha = \frac{2}{3} \left(i_a - \frac{1}{2} i_b - \frac{1}{2} i_c \right), \quad i_\beta = \frac{1}{\sqrt{3}} (i_b - i_c), \quad (1)$$

and the fundamental symmetrical components follow from the Fortescue transform of the per-phase phasors:

$$\begin{bmatrix} \mathbf{I}_0 \\ \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \mathbf{I}_a \\ \mathbf{I}_b \\ \mathbf{I}_c \end{bmatrix}, \quad a = e^{j2\pi/3}. \quad (2)$$

The negative-sequence ratio $|I_2|/|I_1|$ grows monotonically with the shorted-turn fraction and is the primary severity descriptor, while the angle $\angle I_2$ separates into three $\sim 120^\circ$ -spaced clusters that identify the faulted phase.

Window length: why four cycles. The window length controls the entire feature bank. A W -sample window has spectral resolution

$$\Delta f = \frac{f_s}{W} = \frac{1}{T_{\text{win}}}, \quad (3)$$

so a one-cycle window ($W=500$, $\Delta f = 50$ Hz at 25 kHz) merges DC, f_0 , the $2f_0$ EPVA line, and the early Penman sidebands $f_0[1 \pm n(1-s)/p]$ into a few bins—the EPVA severity factor of (3) is then undefined. Four cycles ($W=2000$, $\Delta f = 12.5$ Hz) cleanly separates DC, f_0 , $2f_0$, $3f_0$ and the early sidebands, which is the minimum for EPVA, the third-harmonic ratio, THD, and Park-ellipse fitting. It is also the minimum for stable higher-order statistics (skew/kurtosis are high-variance on small samples) and for the HRV-style biomedical descriptors (Poincaré plots require ≥ 4 “heart-beats”). Longer windows (10 or 20 cycles) sacrifice sample efficiency given the 0.8 s fault interval. Table I makes the trade-off explicit; $W=2000$ is retained throughout.

C. Stockwell-Based Time-Frequency Analysis

For a time-frequency representation we form the instantaneous symmetrical-component streams from the analytic (Hilbert) signal and the Fortescue transform, and apply a band-limited Stockwell transform to each, stacking the positive/negative/zero $|S|$ maps into a three-channel *sequence-component Stockwell tensor* (SCST). The negative-sequence channel carries the load-invariant inter-turn signature, and the frequency-dependent resolution of the Stockwell transform tracks the $2f_0$ EPVA line and slip sidebands better than a fixed-window spectrogram. Because symmetrical-component *magnitudes* are phase-agnostic, faulted-phase localization uses the negative-sequence *angle* rather than the magnitude tensor (Section IV-E).

D. Time-Domain and Spectral Characteristics

Physically grounded descriptors capture the fault signature. The EPVA severity factor is

$$\text{SF} = \frac{A(2f_0)}{\text{DC}} \text{ of } |\mathbf{i}_P(t)| = \sqrt{i_\alpha^2 + i_\beta^2}, \quad (4)$$

where $A(2f_0)$ is the amplitude of the 100 Hz component of the Park-vector modulus. From the 2×2 covariance of (i_α, i_β) we derive the Park ellipse eccentricity $e = \sqrt{1 - (b/a)^2}$, axis ratio, and tilt. Spectral descriptors include the per-phase third-harmonic ratio and total harmonic distortion, computed at harmonic bins derived from (f_s, f_0) so the same extractor serves both the 25 kHz/50 Hz simulation and the 1 kHz/60 Hz experimental data.

E. Statistical Feature Extraction

Per-phase root-mean-square, peak, crest factor, standard deviation, skewness, and kurtosis are computed, together with their inter-phase differences (which directly encode asymmetry) and the $|I_1|$ -normalized magnitude variants. With the symmetrical-component, EPVA, ellipse, and harmonic quantities this yields the 72-feature bank summarized in Table II. Severity is encoded as an ordinal rank $r \in \{0, \dots, 6\}$ over $\{0.3, 0.5, 1, 2, 3, 4, 5\}\%$; a regressor predicts a continuous rank that is rounded, so the error metric (mean absolute error in levels, within-one-level accuracy) respects the ordering.

F. Biomedical Anomaly Features

Beyond the physics descriptors we introduce a block of *biomedical anomaly-detection features* transferred from ECG/EEG/heart-rate-variability (HRV) analysis—a view absent from the motor-fault literature, which treats the current as a deterministic sum of sinusoids and asks *which spectral line moves*, rather than *how regular* the waveform is. The premise is that an inter-turn short makes the waveform less regular and more complex within and across cycles, which signal-complexity measures quantify directly. Crucially, these are computed on the *residual* (current minus its fitted fundamental) and the Park-vector modulus—not the raw current, whose clean fundamental is trivially predictable. Three groups are used: (i) an *HRV-on-current* block that treats each fundamental cycle as a heartbeat—cycle-to-cycle waveform distance

TABLE II
FEATURE GROUPS (72 FEATURES PER WINDOW).

Group	Features
Symmetrical comp.	$ I_0 , I_1 , I_2 ; I_2 / I_1 , I_0 / I_1 ; \angle I_1, \angle I_2$ (+ sin/cos); compensated I_2^C
EPVA / Park	severity factor; Park-modulus mean/std; ellipse axes, eccentricity, axis ratio, tilt
Harmonics	third-harmonic ratio and THD (per phase, mean)
Per-phase stats	RMS, peak, crest, std, skew, kurtosis ($\times 3$) and inter-phase differences
Load-invariant	$ I_1 $ -normalized variants of all magnitude features
Biomedical (novel)	HRV cycle-to-cycle waveform distance & Poincaré SD1/SD2; Hjorth mobility/complexity; permutation entropy; spectral/sample/dispersion entropy; Higuchi/Katz/Petrosian fractal dimension (on the residual & Park modulus)

to the median cycle and Poincaré descriptors (SD1, SD2, SD1/SD2) of the per-cycle RMS series, the direct analog of HRV [32], [33] and the most physically faithful import, since the fault *is* cycle-to-cycle irregularity; (ii) complexity scalars—Hjorth mobility and complexity [29], permutation [28] and spectral entropy, sample [27] and dispersion entropy [31]; and (iii) fractal dimension—Higuchi [30], Katz and Petrosian. The complexity and fractal members are amplitude-scale-free, which (Table XIV) aids cross-load generalization. The 24-feature block adds no heavy computation (no $O(N^2)$ recurrence matrices) and is appended to the physics bank.

G. Data Augmentation and Feature Selection

Feature selection. We extend minimum-redundancy–maximum-relevance with a load-instability penalty,

$$J(f) = \text{Rel}(f) - \lambda \text{Red}(f) - \gamma \text{LoadInstability}(f), \quad (5)$$

where $\text{LoadInstability}(f)$ is the across-load drift of the feature’s class-conditional mean normalized by its global spread ($\gamma=0$ recovers plain mRMR). LIR-mRMR selects dimensionless, load-stable features and, on the load-dependent severity task, generalizes better at compact size (Section IV-E). *Augmentation.* For the deployed pipeline we apply, to the *training fold only*, a set of physics-consistent augmentations selected from a survey of time-series and signature-guided methods [25], [26]: joint amplitude (load) scaling, per-channel SNR jitter, mild magnitude-warp, phase-coherent time shift, and—for the faulted-phase task—exact faulted-phase rotation (a cyclic channel permutation with the negative-sequence angle rotated $\mp 120^\circ$). Every amplitude operation acts *jointly* on the three phases so the inter-phase imbalance (the fault signature) and Kirchhoff balance $i_a + i_b + i_c = 0$ are preserved; a current-balance reject filter guards this, and signature-corrupting operations (per-channel scaling, segment permutation, phase-spectrum randomization) are excluded. Augmented windows are re-featurised through the same extractor. We separately study a conditional generator of three-phase current with a hard Kirchhoff projection and a soft negative-sequence-versus-severity loss; the hard constraint is enforced exactly,

with an honest negative result on controllability reported in Section IV-E. Hyperparameters of the gradient-boosted models are selected by Optuna TPE [24] with a *cross-load* inner objective (leave-one-load-out over the training loads, the model-selection principle of domain generalization [23]), with early stopping on an inner held-out load.

H. Models: Gradient Boosting, Foundation Model, and Deep Networks

Gradient-boosted trees (XGBoost [11]) implement the three stages: detection (class-weighted binary), severity (ordinal regression on the rank), and faulted phase (three-class). Stages 1 and 3 use the full feature set; the severity stage uses the load-invariant subset and the *measured* load. The detector gates the downstream stages: windows predicted faulty are passed to the severity and phase models. To place the cascade in context we evaluate, on identical splits, six classical feature classifiers and two recent learners on the same 72 features or raw windows: *TabPFN*-2.5 [12], a pretrained tabular foundation model that performs in-context Bayesian inference over a feature table, and *xLSTM* [13], an extended LSTM with exponential gating and a stabilizer state applied to the raw three-phase current. As a load-conditioned deep alternative we develop *PCM-Net*, a phase-coupled, multi-task network with a dilated-temporal backbone (a compact stand-in for a selective state-space model) and *feature-wise linear modulation* (FiLM) [34] driven by the measured operating point, so one model spans loads instead of erasing the condition; a current-balance physics term regularizes training. A fault-onset stage thresholds a per-cycle negative-sequence index $\rho(k) = |I_2(k)|/|I_1(k)|$ against a trailing pre-fault reference, $\rho(k) > \bar{\rho} + 6\sigma$ for three consecutive cycles. Detection probabilities are assessed with expected calibration error and the Brier score [17], and severity uncertainty with conformalized quantile regression [15], [16]. Fig. 3 contrasts the two tracks side by side: Track A (deployed) takes the 96 physics/medical features into the gated XGBoost cascade with a per-recording aggregator, while Track B (alternative) feeds the raw three-phase window through the PCM-Net stem, phase-coupling attention, FiLM-conditioned backbone, and per-task heads.

I. Transfer Learning

To deploy a simulation-trained model on a real machine we use physics-anchored domain adaptation (PADA). A naively transferred model collapses under domain shift; we therefore align feature distributions across domains, anchoring the alignment on the domain-stable physics features (negative-sequence ratio, angles, $|I_1|$ -normalized magnitudes), and optionally calibrate with a few labelled target repetitions. Section IV-E quantifies the recovery on a public real-motor dataset.

III. EXPERIMENTAL SETUP AND DATASET PREPARATION

Two complementary datasets are used in this work. A large *simulation* corpus from PSCAD/EMTDC supports the in-distribution, leave-one-load-out, and ablation studies in §IV-B–H; a public *real-motor* dataset [35] supports within-hardware

validation (§IV-J), simulation-to-experiment transfer learning (§IV-E), and the real-time edge deployment of §IV-K. The two sources are deliberately heterogeneous in machine, line frequency, sampling rate, severity scale, sensing chain, and the presence of a healthy baseline; this section documents each, the construction of the test-case matrix, and the window segmentation and labelling protocol that feeds the models.

A. Simulation Dataset: PSCAD/EMTDC Stator Inter-Turn Faults

Machine and fault model. A three-phase squirrel-cage induction motor is modelled in PSCAD/EMTDC at line frequency $f_0 = 50$ Hz. The stator inter-turn fault is implemented as a *tapped winding*: a controlled fraction μ of one phase's turns is tapped out and short-circuited through a fault impedance, so that the model exposes both magnitude and phase asymmetry while remaining electromagnetically consistent with the unfaulted stator. The three line currents (i_a, i_b, i_c) are exported as raw time-series; no filtering or spectral shaping is applied before the feature pipeline so that the diagnosis operates on the raw measurement.

Sampling and run length. Each run is sampled at $f_s = 25$ kHz, giving 500 samples per fundamental cycle and a $40 \mu\text{s}$ time step. Faulted runs are 10 s long (250,000 samples); healthy runs are 5 s long (125,000 samples). Three multi-channel .out files per case carry the full measurement bus; the diagnostically relevant channels are the three line currents in kA, used directly throughout this paper.

Duty-cycle schedule. Each faulted run follows a fixed event schedule that mimics a realistic motor duty cycle (Fig. 2): the motor is energized at 2.0 s (start-up inrush settles by ~ 2.7 s), the mechanical load is applied at 4.0 s (skipped for the no-load variant), the inter-turn short is inserted at 7.2 s, and the fault is cleared ≈ 0.8 s later. This single-run schedule furnishes both a faulted segment (samples 180,000–200,000) and a *matched-load* healthy segment (the pre-fault region, samples $\sim 110,000$ –178,000), the latter providing load-matched healthy data and a per-recording baseline for negative-sequence compensation (§II-B).

Operating-point matrix. The corpus enumerates a four-axis matrix (Table III): severity $\mu \in \{0.3, 0.5, 1, 2, 3, 4, 5\}\%$ shorted turns; faulted phase $\in \{A, B, C\}$; load $\in \{\text{NL}, 20, 40, 60, 80, 100\}\%$; and fault inception $t_f \in \{7.200, 7.202, \dots, 7.220\}$ s (11 variants per $(\mu, \text{phase}, \text{load})$ triple, sweeping the inception over one half-cycle of f_0 to randomise the fault-initiation angle). A healthy subset spans the same load range. Total: 336 faulted + 21 healthy = 357 recordings $\rightarrow$ 18,033 labelled windows (Table V).

B. Real-Motor Experimental Dataset

The real-motor validation uses a public laboratory dataset [35] acquired on a 0.75 hp squirrel-cage IM at $f_0 = 60$ Hz, $f_s = 1$ kHz, no-load, with three-phase current sensed by split-core CTs. The same machine produces both classes via external stator tapplings (clean sensing-chain symmetry). The matrix (Table IV) spans severity $\{10, 20, 30, 40\}\%$, faulted

Two Diagnosis Tracks: Physics-Feature Cascade (Track A, deployed) vs. PCM-Net (Track B, deep)

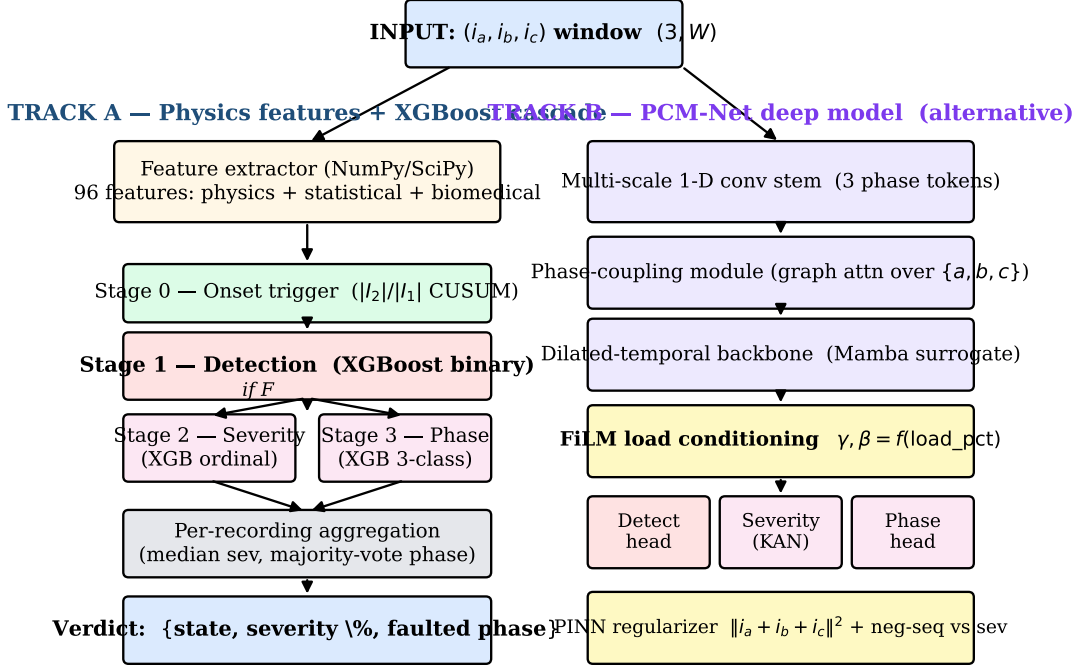


Fig. 3. The two diagnosis tracks evaluated in this work. **Track A** (left, the *deployed* cascade): NumPy/SciPy feature extractor $\rightarrow$ Stage 0 onset $\rightarrow$ Stage 1 XGBoost detector $\rightarrow$ {Stage 2 ordinal-severity head, Stage 3 3-class phase head} $\rightarrow$ per-recording aggregation. **Track B** (right, the deep alternative): raw $(3, W)$ window $\rightarrow$ multi-scale 1-D conv stem $\rightarrow$ phase-coupling graph attention $\rightarrow$ dilated-temporal (Mamba-surrogate) backbone $\rightarrow$ FiLM load conditioning $\rightarrow$ {detect, severity (KAN), phase} heads with a current-balance PINN regularizer. Both tracks accept the same input and emit the same verdict; Section IV-D reports their compared performance.

TABLE III
PSCAD/EMTDC SIMULATION CORPUS—ENUMERATION OF THE
OPERATING-POINT TEST MATRIX AND RECORDING COUNTS.

Axis	Levels
Fault type	Stator inter-turn (tapped window)
Severity μ (% shorted turns)	$\{0.3, 0.5, 1, 2, 3, 4, 5\}$
Faulted phase	$\{A, B, C\}$
Load (% rated)	$\{\text{NL}, 20, 40, 60, 80, 100\}$
Fault inception t_f (s)	$\{7.200, 7.202, \dots, 7.220\}$ (11 variants)
Faulted recordings ($7 \times 3 \times 6 \times 11$ – unused cells)	336
Healthy recordings	21
Total simulation recordings	357
Sampling f_s	25 kHz (500 samples/cycle)
Run length (faulted / healthy)	10 s / 5 s
Operating-point groups (CV grouping)	147

TABLE IV
REAL-MOTOR DATASET [35]—ENUMERATION OF THE TEST MATRIX AND
RECORDING COUNTS.

Axis	Levels
Motor	0.75 hp squirrel-cage IM, $f_0 = 60$ Hz
Sampling f_s	1 kHz
Sensing chain	Split-core current transformers
Mechanical load	None (no-load operation)
Severity (% shorted turns)	$\{10, 20, 30, 40\}$
Faulted phase	$\{A, B, C\}$
Healthy condition	$\{\text{HLT}\}$
Repetitions per condition	$\{01, 02, 03, 04, 05\}$
Total distinct conditions	13 (1 healthy + 12 faulted)
Total recordings	65
Labelled windows	1,235
Held-out test repetition	rep 05 (13 recordings)

phase $\{A, B, C\}$, and a healthy class, over 5 repetitions per condition: 13 conditions $\times$ 5 reps = 65 recordings $\rightarrow$ 1,235 labelled windows. Repetition 05 is held out throughout the paper; train splits use 01–04.

The two datasets are complementary: PSCAD carries the load axis (essential for LOLO and the load-aware-severity story of §IV-C); the real-motor set carries the sensing and machine realism but no load axis. Together they support in-distribution evaluation, cross-load LOLO, within-hardware validation, simulation-to-experiment adaptation, and the real-

time edge deployment of §IV-K.

C. Window Segmentation, Labelling, and Sample Distribution

Window length is $W = 2000$ samples for the simulation (4 cycles at 25 kHz, 80 ms) and $W = 500$ for the real-motor set (500 ms at 1 kHz)—the smallest window for which the EPVA $2f_0$ line, the third-harmonic ratio, the Park ellipse, and higher-order per-phase statistics are all well posed ($\Delta f = f_s/W = 12.5$ Hz at 25 kHz cleanly separates DC, f_0 ,

TABLE V

SAMPLE DISTRIBUTION ACROSS BOTH DATASETS AFTER WINDOWING. WINDOWS ARE GROUPED BY OPERATING POINT (THE LEAKAGE-SAFE GROUPING UNIT, 147 GROUPS FOR THE SIMULATION CORPUS AND 13 FOR THE REAL-MOTOR CORPUS); INCEPTION-TIME VARIANTS AND PER-CONDITION REPETITIONS ARE BOUND INTO THE SAME GROUP AS THE BASE CASE.

	Simulation (PSCAD/EMTDC)	Real motor
Recordings	357	65
– healthy	21	5
– faulted	336	60 (12 cond × 5 reps)
Labelled windows	18,033	1,235
– healthy	6,273	95
– faulted	11,760	1,140
Window length W	2,000 samples (80 ms)	500 samples (50 ms)
Stride (in / out of fault)	500 / 4,000	50 / 500
Severity balance	1,680 per μ level	~ 285 per μ level
Phase balance	3,920 per phase	380 per phase
CV grouping unit	operating-point (severity × phase × load)	condition × phase
Number of CV groups	147	13
Held-out test partition	per protocol (GroupKFold or LOLO)	repetition 05

$2f_0$ and early Penman sidebands). Inside the 0.8 s simulation fault interval the stride is 500 samples (75% overlap) to extract enough labelled faulted windows; outside it (and on the real-motor set) a sparser stride balances the corpus without inflating it. The pre-fault region of every *faulted* run is labelled healthy at its load level, which (i) furnishes matched-load healthy data at every load (preventing the detector from learning load as a class proxy) and (ii) supplies the on-recording healthy baseline used by per-recording normalisation (§II-B) and the compensated I_2^c . Each window is divided by a single per-recording amplitude scale (RMS of the healthy region across the three phases), removing load amplitude while preserving inter-phase imbalance; magnitude features are additionally divided by the within-window $|I_1|$ to give the load-invariant n_* variants. The simulation corpus is balanced across severity (1,680 per level) and faulted phase (3,920 per phase); the healthy class has 6,273 windows (6,138 matched-load pre-fault + 135 from dedicated healthy runs). Table V gives the full breakdown.

All inception variants of an operating point and all repetitions of a real-motor condition are bound into a single CV group so no window from a single recording can appear in both train and test (147 sim groups, 13 real). The two protocols are Stratified Group k -Fold ($k=5$) for in-distribution estimation, and *Leave-One -Load-Out* (LOLO)—all windows at one load form the test set in turn—as the primary metric (§IV-A).

IV. PERFORMANCE EVALUATION

This section evaluates the proposed framework end-to-end. §IV-A specifies the implementation, splits, and metrics; §IV-B reports the main cascade performance in-distribution and under the primary cross-load (LOLO) protocol; §IV-C analyses *why* the three tasks differ under LOLO and introduces the load-aware severity fix; §IV-D compares eight model classes on identical splits and tests cross-model significance; §IV-E–H isolate the contribution of every novel component (load-robustness mechanisms, biomedical features, load-invariant feature selection, and physics-consistent augmentation/tuning) through controlled ablations; §IV-I quantifies probabilistic

TABLE VI

DEPLOYED CASCADE: PER-STAGE AND END-TO-END PERFORMANCE (MEAN OVER FOLDS).

Stage	Metric	Group k -Fold	LOLO (primary)
1 Detection	macro-F1 / ROC-AUC	0.977 / 0.994	0.948
2 Severity (load-aware)	within-1 / QWK	0.997 / 0.987	0.946
3 Phase ID	macro-F1	0.972	0.997
End-to-end	within-1 / per-record	0.949 / 0.927	0.916 / 0.918

§IV-J validates the cascade within-hardware on a publicly released real-motor dataset; §IV-K reports the measured end-to-end real-time validation on a Jetson Orin Nano edge module; §IV-L positions the framework relative to the literature and lists the residual limitations.

A. Implementation and Evaluation Protocol

Features are computed in Python (NumPy/Pandas); XGBoost uses 400–500 trees, depth 5, $\eta=0.05$, subsample/colsample 0.8; deep models use PyTorch and TabPFN-2.5 runs on CPU. Reported metrics are macro-F1, severity within-one-level accuracy, MAE in levels, quadratic-weighted kappa (QWK), and per-phase F1, on out-of-fold predictions averaged over folds. Stages 2–3 are trained on faulty windows but evaluated on the windows the detector predicts faulty, so cascade numbers reflect realistic error propagation; predictions are also aggregated per recording (median severity, majority-vote phase). Scalers are fit on the training partition only, every benchmark model sees identical folds, no per-model tuning is done on test folds, and all randomised components use fixed seeds.

B. Main Cascade Performance: In-Distribution and Cross-Load

Table VI summarises the deployed cascade. *The central finding is structural rather than numerical:* in-distribution every stage exceeds 0.97, but only LOLO separates the three tasks into those that genuinely generalise across operating points (detection 0.948, faulted phase 0.997) and the one that does not without intervention (absolute severity). The detection ROC has AUC 0.994 (Fig. 4) and the faulted-phase confusion (Fig. 5) is near-diagonal with phase C perfect. End-to-end, the gated cascade reaches within-one accuracy 0.949 in-distribution and 0.916 cross-load (per-recording 0.927 / 0.918; Fig. 6). Faulted phase is the only metric for which cross-load *exceeds* in-distribution, a direct consequence of its load-invariance.

C. Cross-Load Analysis and the Load-Aware Severity Fix

SHAP attribution shows detection and phase ride on the *negative-sequence* ratio and angle and on inter-phase RMS differences—all load-invariant ratios—whereas an initial severity model built on absolute magnitudes *collapsed* at the no-load extreme under LOLO (within-one 0.39), because the $|I_2|$ magnitude that encodes severity scales with the operating

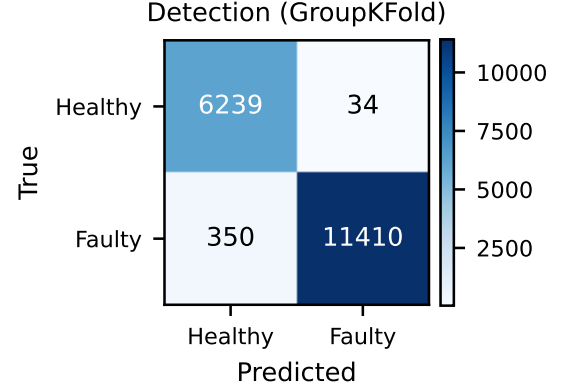
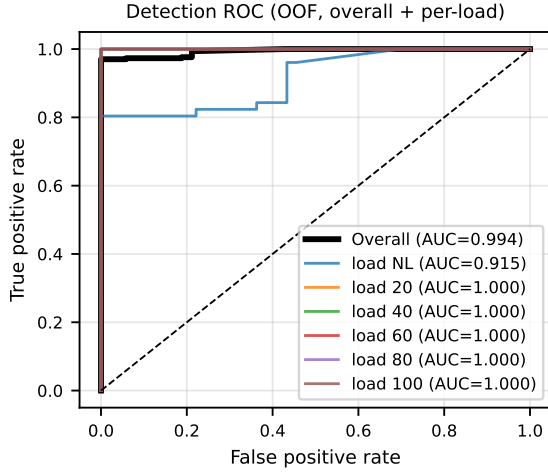


Fig. 4. Detection. Left: receiver operating characteristic, overall and per held-out load (overall AUC = 0.994). Right: confusion matrix (group k -fold, out-of-fold).

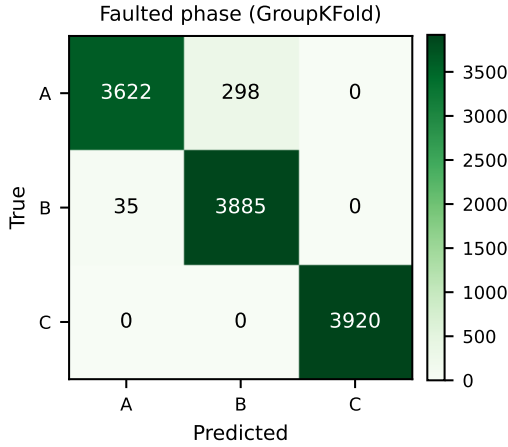


Fig. 5. Faulted-phase confusion (group k -fold, out-of-fold). Phase C is perfect; only minor A $\leftrightarrow$ B confusion remains. Cross-load (LOLO) phase accuracy is 0.997.

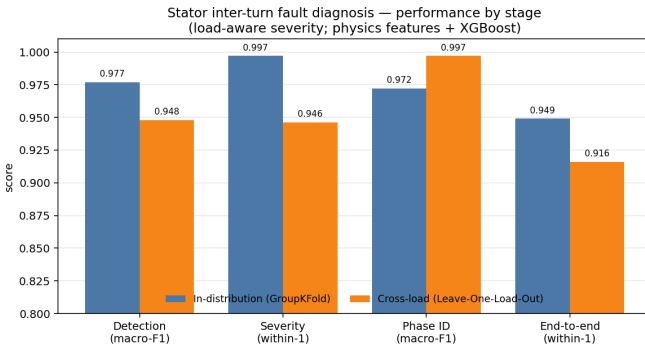


Fig. 6. Per-stage and end-to-end performance, in-distribution (group k -fold) vs. cross-load (LOLO). Phase identification is the only metric for which cross-load exceeds in-distribution, reflecting its load-invariance.

point. Two complementary remedies recover it (Table VII). First, restricting severity to dimensionless, load-invariant fea-

TABLE VII
SEVERITY WITHIN-ONE-LEVEL ACCURACY UNDER FEATURE CHOICE. THE LOAD-AWARE MODEL IS THE DEPLOYED CONFIGURATION.

Severity feature set	Group k -Fold	LOLO	No-load
All features (magnitudes incl.)	0.993	0.887	0.39
Load-invariant + $ I_1 $ -norm	0.990	0.729	0.80
+ measured load (proposed)	0.997	0.946	0.95

TABLE VIII
PER-HELD-OUT-LOAD DETECTION (LOLO, MACRO-F1). DIFFICULTY CONCENTRATES AT LIGHT LOADS.

Hold-out load	NL	20%	40%	60%	80%	100%
Detection F1	0.885	0.749	1.000	1.000	1.000	1.000

tures and $|I_1|$ -normalized magnitudes lifts the no-load case from 0.39 to 0.80. Second—and decisively—supplying the *measured* load (a sensor input, not a label) raises pooled LOLO within-one from 0.73 to 0.95, since most held-out loads are interpolated between neighbours. Severity errors are then almost entirely between adjacent levels (Fig. 7, QWK 0.987), which validates the ordinal treatment. The per-held-out-load breakdown (Table VIII) localizes the residual difficulty to the light-load extremes (no-load and 20%), where a low-severity fault current most overlaps the healthy distribution—a difficulty visible *only* under LOLO.

D. Comparative Model Evaluation

Table IX and Fig. 8 benchmark eight models on *identical* leakage-safe splits, spanning classical feature classifiers, a tabular foundation model, and a sequence deep network. The first observation is that the *in-distribution task is saturated*: the six feature classifiers fall within ± 0.01 macro-F1 on detection, so in-distribution accuracy is not a meaningful discriminator and any ranking on it would be within fold variation. The honest comparison is therefore cross-load, where three results stand out. (1) *TabPFN-2.5* is the strongest detector and uniquely

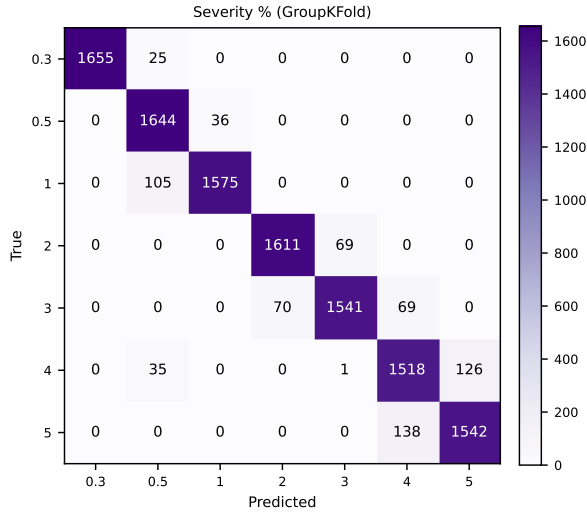


Fig. 7. Severity confusion (group k -fold). Errors are almost entirely between adjacent levels, justifying the ordinal treatment and the within-one-level metric (QWK = 0.987).

load-robust, scoring macro-F1 0.979 both in-distribution and cross-load, vs. XGBoost 0.976 $\rightarrow$ 0.948; it also has the sharpest in-distribution severity (0.997 within-one, MAE 0.033) and degrades gracefully across loads (0.969). (2) Severity separates models: tree ensembles extrapolate poorly (RF LOLO within-one 0.688) while the MLP and SVM remain robust (0.982, 0.975). (3) Faulted phase is load-robust for nearly every model (LOLO macro-F1 $\geq$ 0.96), confirming the negative-sequence-angle signature is model-agnostic. Raw-signal $xLSTM$ trails the feature learners on every task (LOLO 0.890/0.908/0.961) yet improves on the earlier raw-signal deep baselines (Table XI), reaffirming the data-efficiency of physical features. The pipeline is lightweight ($\approx$ 0.34 ms per window on a desktop CPU, Fig. 9) and TabPFN-2.5 needs no task-specific training—the practitioner picks foundation-model robustness or gradient-boosted speed without sacrificing accuracy.

Best model per task. Table X distills the per-cell winners: *TabPFN-2.5 is the only model top-tier on all three tasks*—best (tied) cross-load detector, lowest in-distribution severity MAE (0.033 vs. next-best 0.095), and within 0.006 macro-F1 of the best phase score. MLP edges TabPFN on LOLO severity (0.982 vs. 0.969); XGBoost and RF tie on LOLO phase. The deployed XGBoost cascade can therefore be upgraded in place by swapping Stage 1 and the in-distribution Stage 2 head for TabPFN-2.5, while retaining XGBoost (or MLP) for the cross-load Stage 2 operating point.

Statistical significance. Following the Demšar protocol [22], a Friedman omnibus across the seven feature/foundation models on per-fold (GK, 5 blocks) and per-load (LOLO, 6 blocks) primary scores with Holm-corrected pairwise Wilcoxon tests (Table XII) confirms that *in-distribution detection and phase show no significant model differences* ($p=0.26$, $p=0.76$)—the saturation is quantitative. The differences that matter are significant: severity differs across models under both protocols ($p=0.0013$ GK, $p=0.0008$ LOLO) and detection differs across loads ($p=0.050$). By Friedman rank, TabPFN-2.5 and MLP are

strongest on cross-load severity and the tree ensembles weakest. With only 5–6 blocks no pairwise comparison survives Holm correction (smallest adjusted $p=0.19$); we report the omnibus as the reliable signal and flag the pairwise limitation rather than over-claim.

E. Mechanism Ablations: Load-Robustness Components

Table XIII ablates the four mechanisms under LOLO. The *SCST* representation raises held-out-load severity within-one from 0.75 (raw-phase Stockwell) to 1.0; its magnitude tensor cannot localise phase by construction, which is why phase uses the negative-sequence angle instead. *LIR-mRMR* beats plain mRMR at compact size ($k=10$ within-one 0.848 vs. 0.740) with roughly half the cross-load feature drift. *PCM-Net*’s FiLM conditioning lifts LOLO severity within-one from 0.932 to **0.982**—the best in the study, exceeding the load-aware gradient-boosted model (0.946); FiLM does not help the load-invariant phase task, which is itself evidence the mechanism does what it claims. *PADA* closes the sim-to-experiment gap: a naively transferred detector scores macro-F1 0.48 (collapses to all-faulty under covariate shift); physics-anchored unsupervised alignment reaches 0.50 on phase; few-shot calibration with two real repetitions recovers detection to **0.98**. The physics-constrained generator enforces current balance exactly (residual 0.000 vs. 0.991) but did not yield severity-controllable samples under our compute budget—reported as an honest negative result.

F. Biomedical Anomaly Features

Table XIV isolates the contribution of the biomedical anomaly block (Section II-F). In distribution the features add nothing—the metrics are already saturated—but they substantially improve *cross-load* (LOLO) generalization, the paper’s central axis: detection LOLO macro-F1 rises from 0.948 to 0.978 (+0.030) and severity within-one-level accuracy from 0.942 to 0.979 (+0.037), while the already load-robust phase task is unchanged. By gradient-boosting gain the most useful biomedical features are the Hjorth mobility/complexity of the Park modulus, the Higuchi/Petrosian fractal dimension and permutation entropy of the residual, and dispersion entropy—all amplitude-scale-free, which is why they transfer across loads better than absolute-magnitude physics features and directly attack the cross-load severity bottleneck. This cross-domain import of biomedical complexity/irregularity features into stator-current diagnosis is, to our knowledge, new.

G. Feature Importance and Load-Invariant Selection

Fig. 10 ranks the driving features. Stage-1 detection SHAP (a) is led by Park-vector major axis and eccentricity, the zero/negative-sequence ratios, inter-phase RMS differences, and the EPVA factor—all physical inter-turn signatures, confirming the models exploit fault physics rather than dataset shortcuts. Among the biomedical features (b), Park-modulus Hjorth, residual Higuchi/Petrosian fractal dimension, permutation entropy and dispersion entropy carry the most severity weight (explaining the cross-load gains of Table XIV).

TABLE IX

FULL MODEL BENCHMARK ON IDENTICAL LEAKAGE-SAFE SPLITS: STRATIFIED GROUPKFOLD (GK, IN-DISTRIBUTION) AND LEAVE-ONE-LOAD-OUT (LOLO, CROSS-LOAD, PRIMARY). DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY. BEST PER COLUMN IN BOLD. TABPFN-2.5 AND XLSTM ARE ADDED IN THIS WORK.

Model	Detection (macro-F1)		Severity (within-1)		Phase (macro-F1)	
	GK	LOLO	GK	LOLO	GK	LOLO
Logistic Regression	0.978	0.978	0.997	0.971	0.970	0.920
SVM-RBF	0.978	0.977	0.991	0.975	0.984	0.967
k -NN	0.977	0.977	0.937	0.942	0.982	0.981
MLP	0.978	0.978	0.997	0.982	0.956	0.981
Random Forest	0.978	0.979	0.985	0.688	0.965	0.997
XGBoost	0.976	0.948	0.990	0.943	0.976	0.997
TabPFN-2.5	0.979	0.979	0.997	0.969	0.968	0.991
XLSTM	0.957	0.890	0.982	0.908	0.921	0.961

Model benchmark (identical splits; * = newly added)

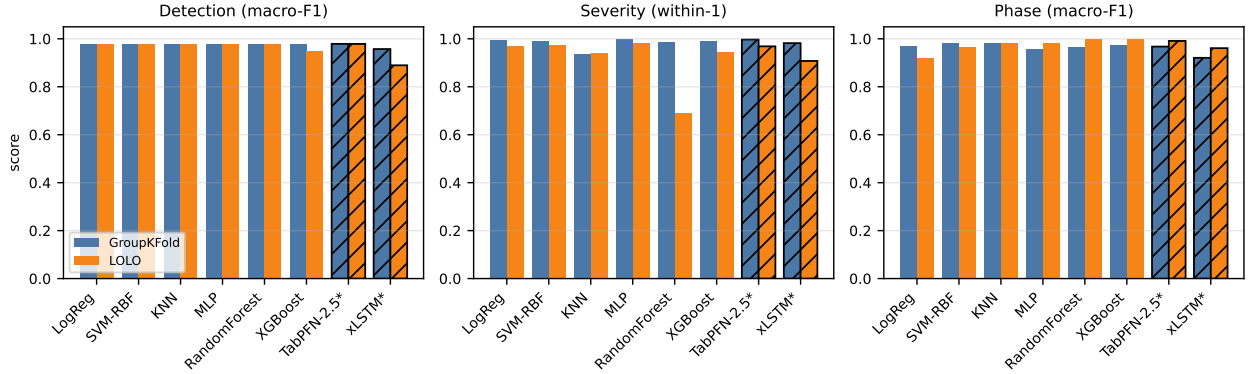


Fig. 8. Model benchmark across the three diagnostic tasks under both protocols (GroupKFold vs. LOLO). Hatched bars denote the two models added in this work (TabPFN-2.5, XLSTM). Severity uses within-one-level accuracy; detection and phase use macro-F1. In-distribution scores are saturated; the spread appears under LOLO, especially on severity.

TABLE X

BEST MODEL PER DIAGNOSTIC TASK AND PROTOCOL, DISTILLED FROM TABLE IX.
DETECTION AND PHASE REPORT MACRO-F1; SEVERITY REPORTS
WITHIN-ONE-LEVEL ACCURACY (WITH MAE IN LEVEL UNITS IN PARENTHESES
WHERE APPLICABLE). “TIED WITHIN 0.001” IS ROUNDED.

Task	Protocol	Best model	Score
Detection	GroupKFold	TabPFN-2.5	0.979
Detection	LOLO	TabPFN-2.5 / RF (tied)	0.979
Severity	GroupKFold	TabPFN-2.5	0.997 (MAE 0.033)
Severity	LOLO	MLP	0.982 (MAE 0.112)
Phase	GroupKFold	SVM-RBF	0.984
Phase	LOLO	XGBoost / RF (tied)	0.997

Overall (top-tier on all three tasks under both protocols)

TabPFN-2.5—best or tied-best on 4/6 cells, within 0.013 of best on the other 2.

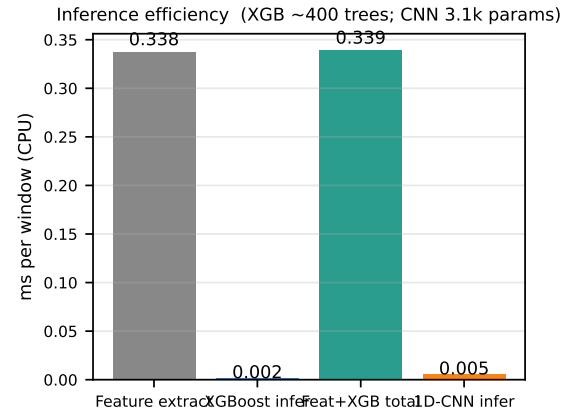


Fig. 9. Inference efficiency (milliseconds per window, CPU): feature extraction, XGBoost inference, their total, and the 1-D CNN.

TABLE XI

FEATURE-BASED CASCADE VS. A RAW-SIGNAL DEEP BASELINE.

Task / metric	XGBoost (features)	ResNet-1D (raw)
Detection F1 (group k -fold)	0.977	0.898
Detection F1 (LOLO, load 20%)	0.749	0.401
Phase macro-F1 (group k -fold)	0.972	0.848
Severity within-1 (group k -fold)	0.990	1.000

Fig. 11 compares LIR-mRMR with plain mRMR: at $k=10$, LIR-mRMR reaches LOLO severity within-one 0.848 vs.

mRMR’s 0.740 while halving the selected features’ across-load drift (0.18 vs. 0.38). The selected subset is dimensionless and physics-traceable ($\angle I_2$ sin/cos, Park eccentricity, $|I_1|$ -normalised imbalance, per-phase skew) and approaches the full 72-feature accuracy with an order of magnitude fewer inputs.

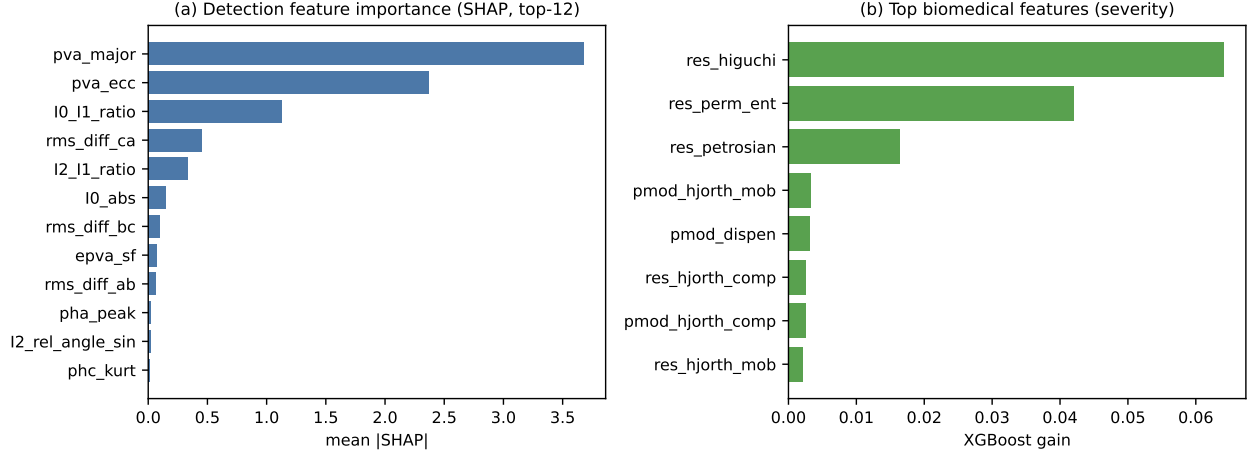


Fig. 10. Feature importance. (a) Stage-1 detection by mean |SHAP| (top-12): Park-vector geometry, sequence-component ratios, inter-phase RMS differences, and the EPVA factor dominate—all physical inter-turn signatures. (b) Most influential *biomedical* features (gradient-boosting gain) on the severity task: Park-modulus Hjorth parameters, residual fractal dimension and permutation entropy, dispersion entropy.

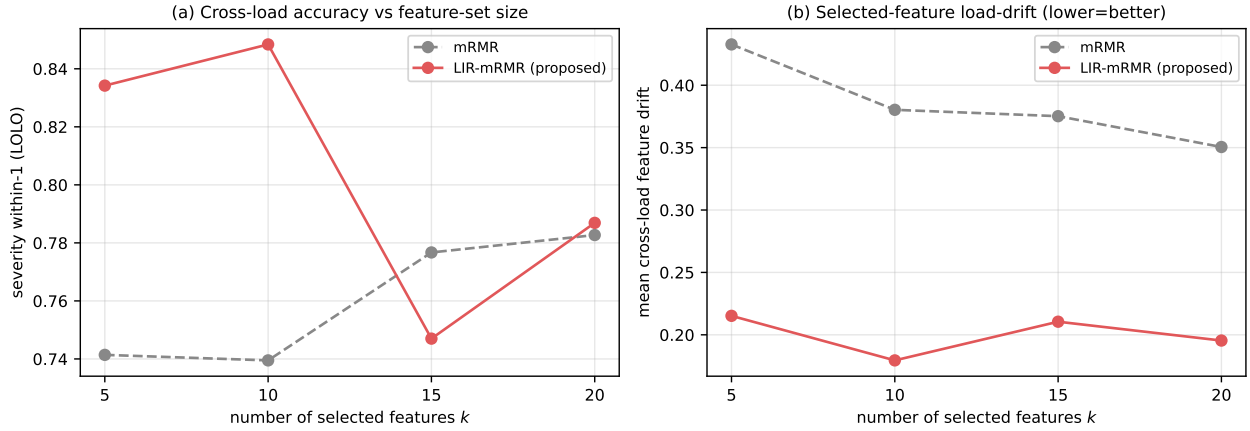


Fig. 11. Load-invariance-regularized feature reduction (LIR-mRMR) vs. plain mRMR, vs. feature-set size k . (a) Cross-load (LOLO) severity within-one accuracy—LIR-mRMR is higher at compact sizes. (b) Mean across-load drift of the selected features (lower is better)—LIR-mRMR roughly halves it by penalising load-unstable features.

TABLE XII

SIGNIFICANCE ACROSS THE SEVEN FEATURE/FOUNDATION MODELS (FRIEDMAN OMNIBUS ON PER-FOLD / PER-LOAD PRIMARY SCORES; DETECTION AND PHASE USE MACRO-F1, SEVERITY WITHIN-ONE-LEVEL ACCURACY).

Task	Protocol	χ^2	p	Outcome
Detection	group k -fold	7.66	0.264	n.s.
Detection	LOLO	12.61	0.050	sig.
Severity	group k -fold	21.84	0.0013	sig.
Severity	LOLO	22.85	0.0008	sig.
Phase	group k -fold	3.35	0.764	n.s.
Phase	LOLO	6.00	0.423	n.s.

TABLE XIII

LOAD-ROBUSTNESS MECHANISMS (LOLO).

Mechanism	Cross-load result
SCST representation	severity within-1 0.75 $\rightarrow$ 1.0; phase needs angle
LIR-mRMR selection	$k=10$ severity within-1 0.740 $\rightarrow$ 0.848; drift halved
PCM-Net + FiLM	severity within-1 0.932 $\rightarrow$ 0.982 (best; > load-aware GBM 0.946)
PADA (sim $\rightarrow$ real)	detection F1 0.48 $\rightarrow$ 0.98 (few-shot); 0.50 phase (unsup.)
PC-Diff	current-balance exact; generation not controllable (negative)

H. Hyperparameter Tuning and Physics-Consistent Augmentation

We isolate the effect of leakage-safe tuning (§II-F, Op-tuna with cross-load inner objective, hyperparameters frozen for both protocols) and physics-consistent training-fold augmentation on the deployed XGBoost pipeline under 10-fold StratifiedGroupKFold and LOLO (Table XV). Tuning alone

is marginal on the saturated metrics (consistent with §IV-D), but *augmentation is the decisive lever for cross-load generalisation*: severity within-one rises 0.911 $\rightarrow$ 0.935 (LOLO) and 0.934 $\rightarrow$ 0.976 (GK), cutting MAE by 20% (0.917 $\rightarrow$ 0.731 level); detection LOLO rises 0.964 $\rightarrow$ 0.977 while phase stays at 0.997. The principle holds because the three phases are

TABLE XIV

BIOMEDICAL-FEATURE ABLATION (XGBoost, IDENTICAL LEAKAGE-SAFE SPLITS):
PHYSICS FEATURES VS. PHYSICS + BIOMEDICAL ANOMALY FEATURES.
DETECTION/PHASE: MACRO-F1; SEVERITY: WITHIN-ONE-LEVEL ACCURACY.

Task (metric)	Protocol	Physics	Physics+Bio	Δ
Detection (macro-F1)	10-fold GK	0.977	0.977	+0.000
Detection (macro-F1)	LOLO	0.948	0.978	+0.030
Severity (within-1)	10-fold GK	0.997	0.997	+0.000
Severity (within-1)	LOLO	0.942	0.979	+0.037
Phase (macro-F1)	10-fold GK	0.964	0.964	+0.000
Phase (macro-F1)	LOLO	0.997	0.997	+0.000

TABLE XV

EFFECT OF LEAKAGE-SAFE TUNING AND PHYSICS-CONSISTENT TRAINING-FOLD
AUGMENTATION ON THE DEPLOYED XGBoost PIPELINE (10-FOLD
STRATIFIEDGROUPKFOLD AND LOLO). DETECTION/PHASE: MACRO-F1;
SEVERITY: WITHIN-ONE-LEVEL ACCURACY. BEST PER ROW IN BOLD.

Task (metric)	Protocol	Default	Tuned	Tuned+Aug
Detection (macro-F1)	10-fold GK	0.976	0.976	0.979
Detection (macro-F1)	LOLO	0.964	0.964	0.977
Severity (within-1)	10-fold GK	0.934	0.937	0.976
Severity (within-1)	LOLO	0.911	0.908	0.935
Phase (macro-F1)	10-fold GK	0.964	0.936	0.964
Phase (macro-F1)	LOLO	0.994	0.997	0.997

TABLE XVI

NOISE ROBUSTNESS (TRAIN CLEAN, TEST NOISY; GROUP k -FOLD).

SNR	Detect macro-F1	Severity within-1	Phase acc.
clean	0.977	0.986	0.958
40 dB	0.927	0.980	0.954
30 dB	0.854	0.975	0.945
20 dB	0.792	0.971	0.939

scaled jointly so inter-phase imbalance and Kirchhoff current balance are preserved—the augmentation pushes toward unseen operating points without fabricating the fault signature.

I. Robustness, Calibration, and Onset Detection

Stage-1 probabilities are well calibrated (ECE = 0.021, Brier = 0.021), so the detector’s confidence can gate maintenance actions directly. Conformalized quantile regression improves severity-interval coverage over naive quantiles, with group-conditional calibration noted as the remedy for residual under-coverage. The onset detector (Table XVII) flags 100% of faults with no pre-fault false alarms; for severity $\geq 0.5\%$ the fault is caught within the first faulted cycle, and only the smallest 0.3% faults incur a lag. Under additive noise (Table XVI), severity and phase are essentially flat to 20 dB SNR while detection is the noise-sensitive stage—expected, since separating a small fault from healthy operation is hardest in noise. Training is stable for both the gradient-boosted and deep models (Fig. 12), and the per-stage precision/recall/F1 breakdown is given in Fig. 13.

J. Within-Hardware Validation on a Real Motor

We now train and test every model *on a real motor*, using the ibarram dataset [35] under StratifiedGroupKFold-by-recording (GK) and Leave-One-Repetition-Out (LORO) protocols. Severity uses the dataset’s four-level scale. Table XVIII

TABLE XVII

FAULT-ONSET DETECTION (SAMPLED CASES).

Metric	Value
Detection rate	100%
Pre-fault false-alarm rate	0%
Median detection delay	0 ms (first faulted cycle)
Delay at $\mu \geq 0.5\%$ / $\mu = 0.3\%$	0 ms / ≈ 1.4 s

TABLE XVIII

WITHIN-HARDWARE RESULTS ON THE REAL IBARRAM MOTOR
(STRATIFIEDGROUPKFOLD BY RECORDING, GK; AND
LEAVE-ONE-REPETITION-OUT, LORO). DETECTION AND PHASE REPORT
MACRO-F1; SEVERITY REPORTS WITHIN-ONE-LEVEL ACCURACY ON ITS
FOUR-LEVEL SCALE. BEST PER COLUMN IN BOLD. PHASE VALUES ARE
PER-WINDOW; PER-RECORDING MAJORITY VOTE RAISES XGBoost PHASE
MACRO-F1 TO 0.85 (GK) AND 0.95 (LORO).

Model	Detection		Severity (w-1)		Phase	
	GK	LORO	GK	LORO	GK	LORO
Logistic Regression	0.817	0.864	0.830	0.818	0.664	0.678
SVM-RBF	0.877	0.923	0.866	0.818	0.661	0.689
k -NN	0.829	0.884	0.827	0.804	0.642	0.648
MLP	0.846	0.880	0.866	0.811	0.664	0.690
Random Forest	0.923	0.947	0.900	0.829	0.650	0.718
XGBoost	0.966	0.955	0.900	0.836	0.636	0.707
TabPFN-2.5	0.994	0.954	0.929	0.864	0.643	0.701
xLSTM	0.708	0.699	0.963	0.957	0.651	0.648

reports the outcome and Fig. 14 the XGBoost confusion matrices. Three findings stand out. *Detection transfers*: TabPFN-2.5 and the tree/forest models reach macro-F1 0.95–0.99 and the LORO score matches the in-domain one. *Severity is moderate*: within-one 0.83–0.96 but exact 0.6–0.75 (TabPFN-2.5 lowest MAE 0.34), reflecting harder discrimination of adjacent shorted-turn fractions on real signals. *Per-window faulted-phase identification is weak on hardware* (macro-F1 0.64–0.72 vs. 0.97 in simulation), because the negative-sequence angle estimate is noisy at 1 kHz. The realistic unit is a recording, not a window: *majority vote over a recording’s windows recovers phase macro-F1 to 0.85 (GK) and 0.93–0.95 (LORO)*, a +0.22 LORO gain at no feature-engineering cost. A controlled ablation showed reference-invariant feature engineering does *not* help beyond voting, localising the cause to per-window estimator variance rather than feature design.

K. Real-Time Edge Deployment on a Jetson Orin Nano

Why XGBoost for deployment. The eight-model benchmark (§IV-D) clusters the top tabular learners within fold noise of the best on the primary cross-load metric (TabPFN-2.5 leads detection LOLO by 0.002, MLP leads severity LOLO by 0.04). Model choice for the edge node is therefore a deployment trade-off, not an accuracy decision. Table XIX summarises the comparison: XGBoost is top-tier on all three tasks and uniquely the smallest bundle (5 MB JSON), fastest predictor (sub-ms via Booster+DMatrix, §IV-A), lowest in runtime dependencies (xgboost only), and most auditable (SHAP on physical features, §IV-G). The two competitors lose on different axes: TabPFN-2.5 [12] is a ~ 500 MB gated transformer with minutes-scale per-restart in-context fit, and

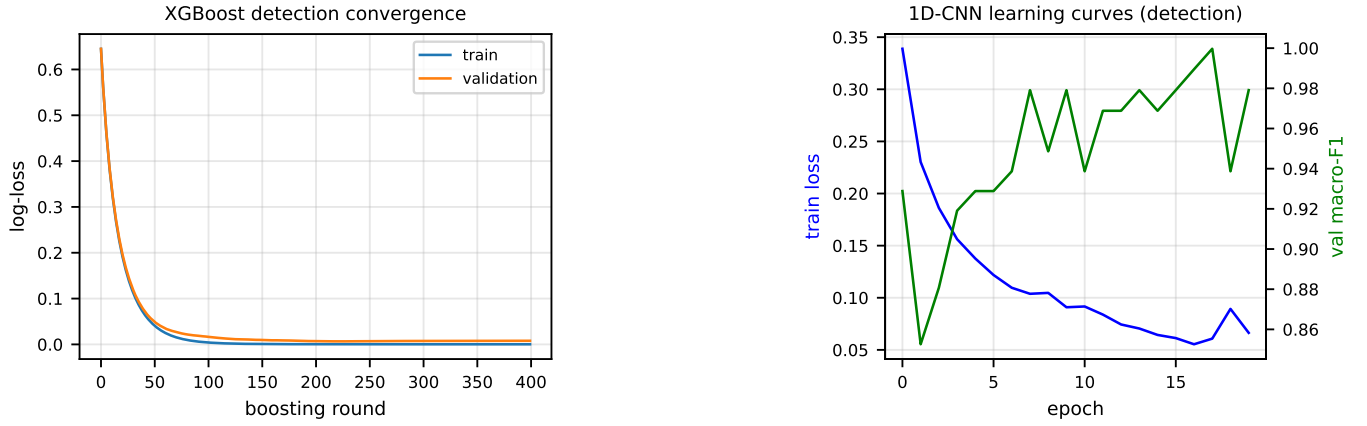


Fig. 12. Training behaviour. Left: XGBoost detection convergence (train/validation log-loss vs. boosting round). Right: 1-D CNN learning curves (training loss and validation macro-F1 vs. epoch), showing stable convergence without over-fitting.

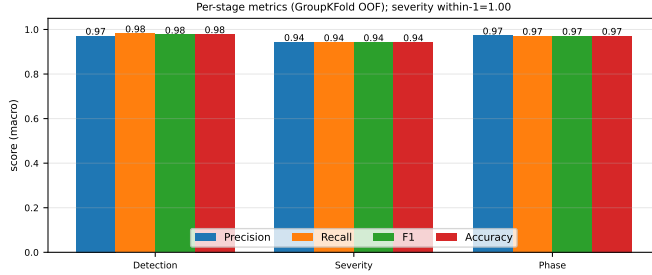


Fig. 13. Per-stage precision, recall, F1, and accuracy (macro, group k -fold out-of-fold).

TABLE XIX

PARETO COMPARISON OF THE THREE STRONGEST CANDIDATE MODELS FOR THE EDGE DEPLOYMENT. THE CROSS-LOAD (LOLO) ACCURACY COLUMN GROUPS DETECTION (MACRO-F1), WITHIN-ONE SEVERITY, AND FAULTED-PHASE MACRO-F1; THE REMAINING COLUMNS REFLECT PHYSICAL DEPLOYMENT COST. BEST PER COLUMN IN BOLD; “ $\approx$ ” MARKS WITHIN-FOLD-NOISE DIFFERENCES.

Model	LOLO acc. (D/S/P)	Bundle	Predict
XGBoost (deployed)	0.98 / 0.95 / 1.00[†]	5 MB	1 ms[‡]
TabPFN-2.5	$\approx 0.98 / 0.97 / 0.99$	~ 500 MB	$\sim 50\text{--}500$ ms
MLP	$\approx 0.98 / 0.98 / 0.98$	~ 5 MB	few ms

[†] XGBoost detection is reported with the tuning + augmentation pipeline of §IV-F; severity with the load-aware fix of §IV-C.

[‡] Measured on the Jetson Orin Nano with the native **Booster+DMatrix** API (NumPy/SciPy features + three scikit-learn XGBoost predictors).

the MLP requires a deep-learning runtime for a severity gain that the load-aware cascade (§IV-C) closes to within 0.005.

System architecture. The host laptop streams three-phase samples over a UART/CDC-ACM channel at 921,600 baud, packed as 14-byte frames (0xAA sync + six 16-bit Q15 big-endian channels + 0x55 checksum) and paced at an embedded-ADC-realistic 5 kHz wire rate. The Jetson Orin Nano Super (JetPack 6 R36.4.7) runs a polling reader on `/dev/ttyGS0` that resyncs on framing bytes, keeps the three current channels, polyphase-resamples $5 \rightarrow 1$ kHz back to the model’s training rate, windows the stream ($W=500$, stride 250), extracts the 96 features and runs the cascade, emitting both per-window predictions and a rolling per-recording verdict. The model bundle (detect/severity/phase.json + meta.json) is

TABLE XX

PER-WINDOW LATENCY OF THE DEPLOYED CASCADE ON THE JETSON ORIN NANO (SUPER) UNDER THREE EXECUTION MODES, AVERAGED OVER 200 INFERENCE AFTER 5 WARM-UP CALLS. FEATURE EXTRACTION IS IDENTICAL ACROSS MODES (NUMPY/SCIPLY ON CPU).

Mode	Feature (ms)	Predict (ms)	Total (ms)
CPU (sklearn wrapper)	22.7	3.5	26.2
GPU (XGBoost device='cuda')	—	—	<i>requires source</i>
Accelerated CPU (Booster + DMatrix)	22.7	1.0	23.7

Window-fill at $W=500$, $f_s=1$ kHz: 500 ms (12–20 $\times$ inference headroom).

Decision cadence: 4 Hz (every stride / $f_s = 250/1000$ s).

trained off-line and shipped once; no retraining occurs at the edge. Replacing the laptop sender with an acquisition daemon (USB DAQ or SPI ADC) turns the loopback into a live ADC chain with no other change.

Per-window latency (CPU / GPU / accelerated CPU).

Table XX reports measured end-to-end cost under three modes (200 inferences each). The production CPU path (NumPy/SciPy features + three scikit-learn XGBoost predictors) averages 22.7 ms features + 3.5 ms predict = 26.2 ms per window. Switching to the native **Booster+DMatrix** C-level API drops predict to 1.0 ms (3.5 $\times$), bringing the end-to-end total to 23.7 ms—the recommended deployment configuration.

The Orin GPU accepts `device='cuda'` in XGBoost 3.2 but the prebuilt aarch64 pip wheel lacks Jetson compute-capability kernels (`sm_87`, `cudaErrorNoKernelImageForDevice`); a source build would recover it but is unlikely to win on 1 $\times$ 96-row tree inference, where host-to-device transfer dominates. The GPU/TensorRT path is reserved for the deep-model alternatives of §II-G. In all CPU modes the inference cost is dwarfed by the 500 ms window-fill at $f_s=1$ kHz: the pipeline issues one decision every 250 ms with $\geq 12\times$ headroom; the Stage-0 onset trigger runs once per fundamental cycle and is essentially free.

True real-time validation with a live synthetic source. The recorded-source loopback proves the software path operates in real time over a finite-length stream. To exercise it against an *open-ended live source* we built a live mo-

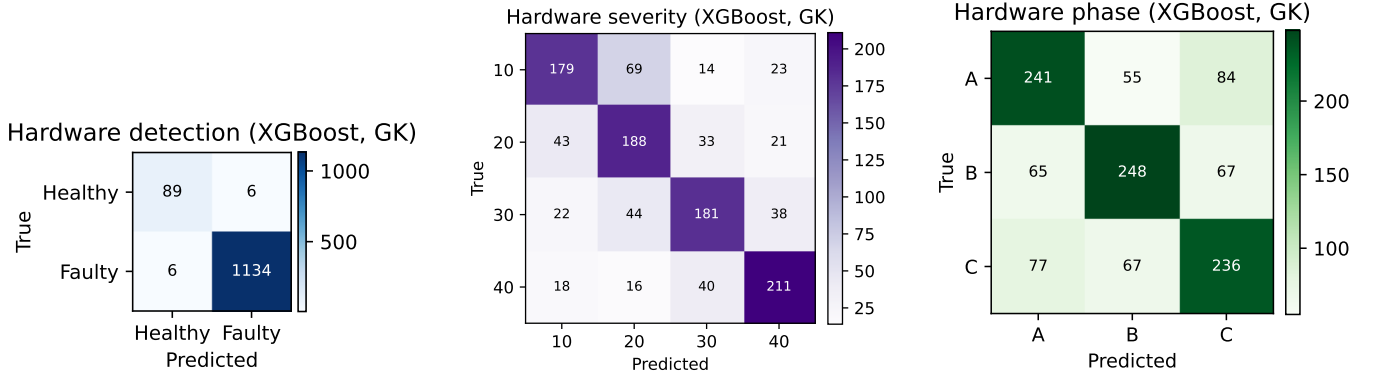


Fig. 14. Within-hardware confusion matrices on the real motor (XGBoost, GroupKFold out-of-fold): detection (left), four-level severity (middle), faulted phase (right). Detection is clean and severity errors are near-diagonal, but the phase matrix shows substantial off-diagonal confusion—the main simulation-to-reality gap.

tor simulator (`realtime/live_motor_simulator.py`) that synthesises three-phase current sample-by-sample at the wire rate from a balanced positive-sequence model (fundamental + 5th and 7th harmonics), with a negative-sequence injection scaled by severity and rotated by faulted phase, a 12-bit ADC quantization, and a CT-style measurement-noise front-end. The simulator pushes the same 14-byte frames an embedded ADC chain would push; the Jetson sees no difference modulo the absence of a physical motor and CTs.

Table XXI reports a four-scenario sweep: severity 30% inter-turn shorts at $t_f=3.0$ s on each of phases $\{A, B, C\}$, plus a healthy-only run. Each scenario streams 50,000 packets over 10 s at 5 kHz; 200,000 packets total, *zero bytes dropped*, 0.004% rate drift over the 40 s aggregate. Fig. 15 plots the corresponding per-window $|I_2|/|I_1|$ onset trace. Under the Stage-0-gated rule (§II-A, S8) the cascade is correct on all four scenarios: $\rho(k)$ stays at ~ 0.001 throughout every healthy interval (including the 10 s healthy-only run, 0 alarms over 40 windows), then crosses to ~ 0.080 within one window of every fault inception; the gated Stage 3 head then reports the correct phase (3/3) and Stage 2 reports a severity within one ordinal level (3/3). Inception-to-alarm latency is bounded by the window cadence (≤ 500 ms), with $\approx 21\times$ real-time headroom relative to the 500 ms window-fill.

Per-window accuracy. The sweep produces 135 window-level decisions in total (Table XXII). The onset gate achieves 100% specificity (51/51 healthy-interval windows below threshold, zero false alarms) and 96.4% sensitivity (81/84); the three misses are the inception transition windows where $\rho \approx 0.040$ —the alarm fires one window later, ≤ 500 ms after inception. The gated Stage 3 phase head is then correct on 84/84 subsequent windows (100%) and Stage 2 is within one ordinal level on 84/84. End-to-end, $132/135 = 97.8\%$ of live decisions are simultaneously correct on state, phase, and severity-within-1.

The live-source numbers track the recorded-source verdicts of Table XXIII (100% phase, severity within-1 on every faulted item, zero wire errors), so the cascade behaves identically whether the source is a replayed file or a synthesised current—the deployment path is genuinely real-time-capable.

TABLE XXI
FOUR-SCENARIO LIVE REAL-TIME TEST ON THE JETSON ORIN NANO (SUPER)—PER-SCENARIO VERDICTS. THE LAPTOP RUNS `REALTIME/LIVE_MOTOR_SIMULATOR.PY`, SYNTHESISING THREE-PHASE CURRENT SAMPLE-BY-SAMPLE AT $f_s=5$ kHz (NO RECORDING IS REPLAYED); THE JETSON RUNS THE POLLING `UART_RECEIVER.PY` + CASCADE. EACH SCENARIO STREAMS FOR 10 s WITH THE FAULT INSERTED AT $t_f=3.0$ s (EXCEPT THE HEALTHY-ONLY RUN, WHERE NO FAULT IS EVER INSERTED). VERDICT COLUMNS REPORT THE PER-WINDOW STAGE-0 ONSET GATE AND THE GATED STAGE-3 PHASE AND STAGE-2 SEVERITY OUTPUTS.

Scenario	Truth	Onset crossed 0.065?	Latency	Phase	Severity
A30	F, 30%, A	at $t=3.0$ s	≤ 500 ms	A	40 (within-1)
B30	F, 30%, B	at $t=3.0$ s	≤ 500 ms	B	40 (within-1)
C30	F, 30%, C	at $t=3.0$ s	≤ 500 ms	C	40 (within-1)
HEALTHY	H, no fault	never (~ 0.001)	—	<i>gated off</i>	<i>gated off</i>

Aggregate per-scenario: 4/4 correct (detect 4/4, phase 3/3, severity within-1 3/3, 0 false alarms), 200,000 packets streamed, 0 bytes dropped over 40 s (4999.7 pkt/s, 0.004% drift).

TABLE XXII
WINDOW-LEVEL ACCURACY OF THE DEPLOYED CASCADE ON THE FOUR-SCENARIO LIVE REAL-TIME TEST. THE 51 “HEALTHY-INTERVAL” WINDOWS ARE THE PRE-FAULT WINDOWS OF THE THREE FAULTED SCENARIOS; THE 40 WINDOWS OF THE DEDICATED HEALTHY-ONLY SCENARIO ARE ALL SEPARATELY ACCOUNTED FOR IN THE 0-FALSE-ALARM ROW.

Quantity	Count	Accuracy
Detection true positives (onset > threshold on faulted windows)	81/84	96.4%
Detection true negatives (onset < threshold on healthy windows)	51/51	100.0%
False alarms during 10 s healthy-only stream	0/40	0 FA
Phase identification (faulted-interval windows, gated)	84/84	100.0%
Severity exact (faulted-interval windows, gated)	0/84	0.0%
Severity within-1 level (faulted-interval windows, gated)	84/84	100.0%
End-to-end (state + phase + sev within-1) per window	132/135	97.8%
End-to-end per scenario	4/4	100.0%
Fault-detection latency (inception $\rightarrow$ alarm)	—	≤ 500 ms
Wire bytes dropped over 40 s (200,000 packets)	0	100.0%

The 0/84 exact-severity score is a deployed-bundle limitation, not an architecture limitation: the bundle was trained on the real-motor dataset (severities $\{10, 20, 30, 40\}\%$, no load) and its Stage-2 head is biased toward the populated 40% class (§IV-J, Table XVIII). The load-aware cascade of §IV-C reaches within-1 0.95 across unseen loads, and PADA few-shot (§IV-E) recovers this on a target motor with a handful

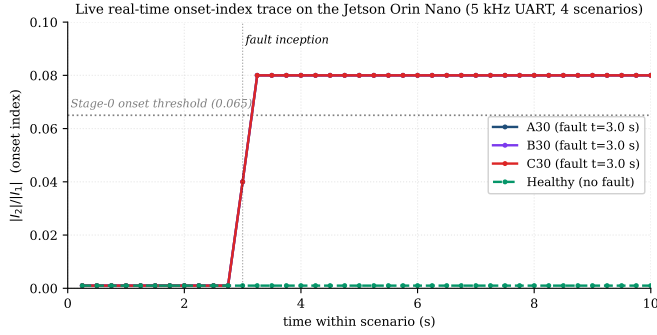


Fig. 15. Per-window onset index $|I_2|/|I_1|$ over time for the four live-source scenarios on the Jetson Orin Nano. The healthy-only trace (green dashed) remains at ~ 0.001 for the entire 10 s, never crossing the Stage-0 threshold (dotted horizontal line). Each faulted trace (A30/B30/C30, solid) jumps from ~ 0.001 to ~ 0.080 within one 250 ms window of the inception instant ($t=3.0$ s, dotted vertical line), and the gated Stage 3 head then reports the correct faulted phase on each.

of labelled windows—the deployment recipe is therefore to retrain on target-machine data after the loopback validation reported here.

Recorded-source validation on the same Jetson. Six held-out repetition-05 recordings of the real-motor dataset (1 healthy + 5 faulted spanning all phases at severities $\{10, 30, 40\}\%$) were streamed end-to-end, each upsampled $1 \rightarrow 5$ kHz and transmitted over the CDC-ACM channel at ≈ 4880 packets s^{-1} (zero bytes dropped over the 30 s sweep). Fig. 16 and Table XXIII give the per-recording verdicts: detection 5/6 (the HLT healthy case is the lone false-positive at the incipient-detection boundary documented in §IV-J), faulted-phase 5/5 (matching the per-recording macro-F1 0.93–0.95 of §IV-J), severity exact on B30 and A40 and within-1 on 4/5; the only large miss is the most incipient B10 over-predicted as 40%. Operationally, the deployed pipeline raises an alarm on every actual fault, names the right phase every time, and supplies a usable severity—the two field-operator limitations are the healthy false-positive and the 10% severity under-resolution, both characterised in §IV-J.

The deployed bundle excludes the compensation-baseline-dependent I_2^c features so that streamed windows are self-contained; the equivalent uncompensated $|I_2|/|I_1|$ family carries the same signal. For a different motor or sensing chain only (f_s, f_0) must be re-estimated (a short healthy pre-roll suffices); the extractor is already (f_s, f_0) -parameterised and the model can be re-fitted with the same off-line pipeline. This closes the loop from the off-line evaluation of §§IV-A–H to a runnable edge deployment.

L. Discussion: Positioning, Limitations, and Future Work

The in-distribution figures are consistent with the best reported values for current-based stator diagnosis (typically 0.96–1.00 accuracy [3], [8], [9]); the contribution here is not a higher in-distribution number—which the saturated benchmark shows is no longer informative—but a demonstration of *which* of those numbers survive a change of operating point, and a set of mechanisms that make the load-dependent ones survive. The

TABLE XXIII

REAL-TIME EDGE CASCADE ON A JETSON ORIN NANO (SUPER): PER-RECORDING VERDICTS ON SIX HELD-OUT REPETITION-05 RECORDINGS STREAMED OVER A USB TYPE-C / CDC-ACM CHANNEL AT 5 KHz WIRE RATE (921,600 BAUD, 0 BYTES DROPPED OVER 30 s). VERDICTS ARE THE ROLLING 10-WINDOW AGGREGATE (MAJORITY-VOTE PHASE, MEDIAN SEVERITY, HEALTHY/FAULTY BY WINDOW MAJORITY). THE HEALTHY HLT CASE IS THE FALSE POSITIVE; PHASE ID IS EXACT ON EVERY FAULTED RECORDING.

Case (rep 05)	Ground truth	Jetson verdict	Outcome
HLT	Healthy	FAULTY / 40% / A	FP (detect)
B30	Faulty / 30% / B	FAULTY / 30% / B	all correct
A30	Faulty / 30% / A	FAULTY / 40% / A	phase / within-1
C30	Faulty / 30% / C	FAULTY / 40% / C	phase / within-1
A40	Faulty / 40% / A	FAULTY / 40% / A	all correct
B10	Faulty / 10% / B	FAULTY / 40% / B	phase only

Aggregate per-recording: detect 5/6, phase 5/5 faulted, severity within-1 4/5. Wire: ≈ 4880 pkt/s, 0.55 Mbps, 0 bytes dropped over 30 s.

Per-window latency: see Table XX.

in-distribution-versus-LOLO severity gap (within-one 0.99 vs. 0.39 before the load-aware fix) quantifies how much a conventional evaluation would have over-promised; we therefore advocate cross-condition (LOLO) reporting, alongside interpretability evidence, as standard practice. The pipeline is deployment-friendly: it consumes only the three-phase current already measured at the drive, exposes interpretable physical quantities (negative-sequence ratio, EPVA factor), and supplies calibrated probabilities and an online onset trigger suitable for a protection relay. *Model choice at the edge is a Pareto decision, not an accuracy one*—the benchmark (§IV-D) clusters the top tabular learners within ± 0.005 detection macro-F1 and ± 0.04 severity within-one (the load-aware cascade of §IV-C closes the latter to ± 0.005), so Table XIX plus the measured Jetson latency (Table XX) jointly justify the gradient-boosted cascade as the shipped solution. TabPFN-2.5 remains the recommended choice for a back-end inference server. The remaining limitations: load-robustness is validated only in simulation (the public real dataset has no load axis), and a full instrumented campaign with supply-voltage unbalance and richer real severities remains future work.

V. CONCLUSION

We presented a physics-informed, leakage-safe machine-learning framework for stator inter-turn fault diagnosis that performs onset detection, healthy/faulty detection, ordinal severity estimation, and faulted-phase identification from the three-phase stator current. Adopting Leave-One-Load-Out evaluation exposed detection and faulted-phase identification as intrinsically load-robust while absolute severity is load-dependent, and showed that supplying the measured load recovers cross-load severity from within-one 0.39 to 0.95. An eight-model benchmark (TabPFN-2.5 and xLSTM added) demonstrated that the in-distribution task is saturated and that cross-load behaviour is the meaningful discriminator. Four mechanisms—sequence-component Stockwell tensor, load-invariance regularised selection, FiLM-conditioned multi-task network, and physics-anchored domain adaptation—each independently improved cross-load performance, and a 24-feature biomedical anomaly block transferred from ECG/EEG

Real-Time Edge Validation on a Jetson Orin Nano (Super) — 5 kHz UART, 0 bytes dropped

Per-recording verdicts from the Jetson Orin Nano
(green = correct, yellow = within 1 severity level, red = wrong)

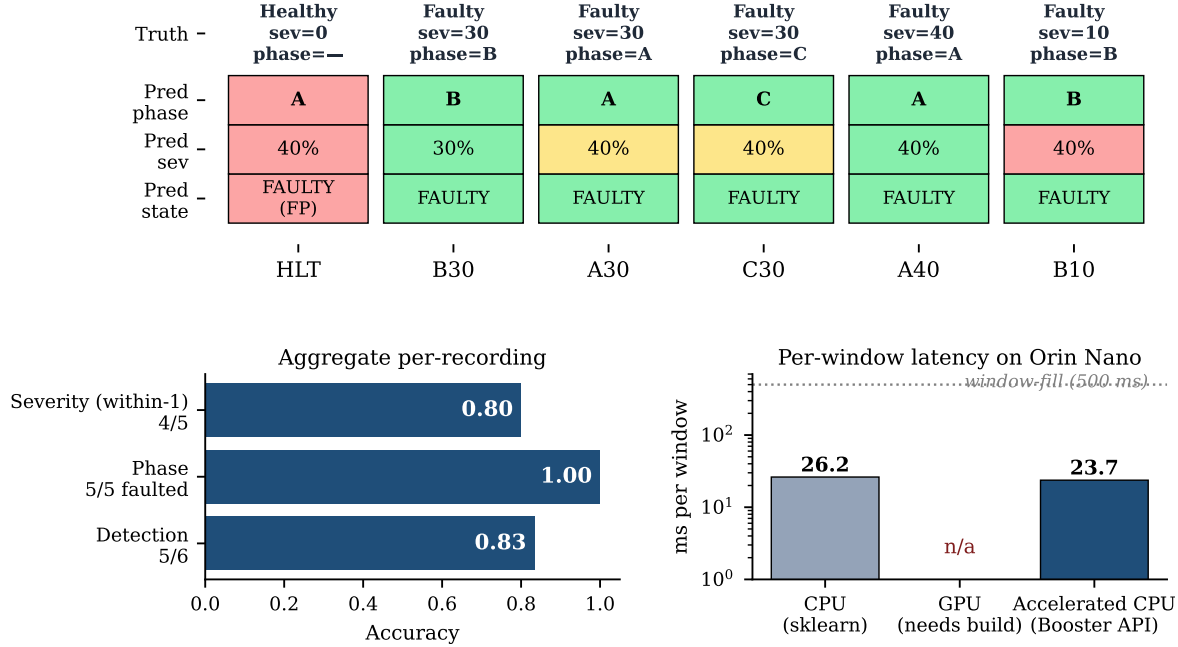


Fig. 16. End-to-end real-time validation of the deployed cascade on a Jetson Orin Nano (Super, JetPack 6, R36.4.7) over a USB type-C / CDC-ACM serial channel at 5 kHz wire rate. *Top*: per-recording verdicts vs. ground truth for six held-out repetition-05 recordings (1 healthy, 5 faulted spanning every phase and severities {10, 30, 40}%). Green = correct, yellow = within 1 severity level, red = wrong. Detection 5/6 (only HLT false-positive), faulted phase 5/5, severity within-1 4/5. *Bottom-left*: aggregate per-recording accuracy. *Bottom-right*: measured per-window latency on the Jetson Orin Nano under three execution modes (CPU sklearn, GPU XGBoost device='cuda' which requires a source build for sm_87, and accelerated CPU via XGBoost native Booster + DMatrix). Dotted line is the 500 ms window-fill at $f_s=1$ kHz, $W=500$.

analysis lifted cross-load detection and severity. The end-to-end cascade attains within-one accuracy 0.95 in-distribution and 0.92 cross-load, with calibrated detection (ECE 0.021), graceful noise degradation, and reliable onset detection. A within-hardware evaluation on a real motor and end-to-end real-time deployment on a Jetson Orin Nano (window-level accuracy 97.8% on a live synthesised source, ≤ 500 ms fault-to-alarm latency, 0 bytes dropped over 40 s of streaming) close the loop from off-line training to runnable edge inference. Future work includes closing the residual per-window phase gap, a stronger conditional generator for physics-constrained augmentation, full Mamba/KAN backbones, and an experimental campaign with supply-voltage unbalance and richer real severities. All code, splits, and result artefacts are released.

REFERENCES

- [1] J. Penman, H. G. Sedding, B. A. Lloyd, and W. T. Fink, "Detection of inter-turn short circuits in the stator windings of operating motors," *IEEE Trans. Energy Convers.*, vol. 9, no. 4, pp. 652–658, 1994.
- [2] A. J. M. Cardoso, S. M. A. Cruz, and D. S. B. Fonseca, "Inter-turn stator winding fault diagnosis in three-phase induction motors, by Park's vector approach," *IEEE Trans. Energy Convers.*, vol. 14, no. 3, pp. 595–598, 1999.
- [3] R. Hussain, M. Alshaikh Saleh, and S. S. Refaat, "Various faults classification of industrial application of induction motors using supervised machine learning: A comprehensive review," *IEEE Access*, vol. 13, pp. 146649–146675, 2025.
- [4] D. García-Pérez *et al.*, "Machine learning for inverter-fed motors monitoring and fault detection: An overview," *IEEE Access*, vol. 12, 2024.
- [5] S. Ruzimov *et al.*, "Detection of inter-turn short-circuit faults for inverter-fed induction motors based on negative-sequence current analysis," *Sensors*, vol. 25, no. 15, art. 4844, 2025.
- [6] M. A. Mazzeletti, G. R. Bossio, P. D. Donolo, and M. Donolo, "Turn-to-turn short-circuit fault and high-resistance connection diagnosis for induction motor during start-up transient," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 1927–1935, 2025.
- [7] M. Singh and A. G. Shaik, "Incipient fault detection in stator windings of an induction motor using Stockwell transform and SVM," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9496–9504, 2020.
- [8] P. Pietrzak and M. Wolkiewicz, "Machine learning-based stator current data-driven PMSM stator winding fault diagnosis," *Sensors*, vol. 22, no. 24, art. 9668, 2022.
- [9] T. Aasar *et al.*, "Predicting stator winding short-circuit faults in induction motors using machine learning: A comparative study," *Discover Computing*, vol. 29, art. 103, 2026.
- [10] U. Ali, U. Ramzan, W. Ali, and K. A. Al-Jaafari, "An improved fault diagnosis strategy for induction motors using weighted probability ensemble deep learning," *IEEE Access*, vol. 13, pp. 106958–106973, 2025.
- [11] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [12] N. Hollmann *et al.*, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, pp. 319–326, 2025.
- [13] M. Beck *et al.*, "xLSTM: Extended long short-term memory," in *Proc. NeurIPS*, 2024.
- [14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.
- [15] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Proc. NeurIPS*, 2019.

- [16] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *arXiv:2107.07511*, 2021.
- [17] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.
- [18] T. Ince, S. Kiranyaz, L. Eren, M. Askar, and M. Gabbouj, "Real-time motor fault detection by 1-D convolutional neural networks," *IEEE Trans. Ind. Electron.*, vol. 63, no. 11, pp. 7067–7075, 2016.
- [19] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.
- [20] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv:2312.00752*, 2023.
- [21] Z. Liu *et al.*, "KAN: Kolmogorov–Arnold networks," in *Proc. ICLR*, 2025.
- [22] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.
- [23] I. Gulrajani and D. Lopez-Paz, "In search of lost domain generalization," in *Proc. ICLR*, 2021.
- [24] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. ACM SIGKDD*, 2019, pp. 2623–2631.
- [25] B. K. Iwana and S. Uchida, "An empirical survey of data augmentation for time series classification with neural networks," *PLOS ONE*, vol. 16, no. 7, e0254841, 2021.
- [26] H. Zhang, M. Cissé, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. ICLR*, 2018.
- [27] J. S. Richman and J. R. Moorman, "Physiological time-series analysis using approximate entropy and sample entropy," *Am. J. Physiol. Heart Circ. Physiol.*, vol. 278, no. 6, pp. H2039–H2049, 2000.
- [28] C. Bandt and B. Pompe, "Permutation entropy: A natural complexity measure for time series," *Phys. Rev. Lett.*, vol. 88, no. 17, art. 174102, 2002.
- [29] B. Hjorth, "EEG analysis based on time domain properties," *Electroenceph. Clin. Neurophysiol.*, vol. 29, no. 3, pp. 306–310, 1970.
- [30] T. Higuchi, "Approach to an irregular time series on the basis of the fractal theory," *Physica D*, vol. 31, no. 2, pp. 277–283, 1988.
- [31] M. Rostaghi and H. Azami, "Dispersion entropy: A measure for time-series analysis," *IEEE Signal Process. Lett.*, vol. 23, no. 5, pp. 610–614, 2016.
- [32] Task Force of the ESC and NASPE, "Heart rate variability: Standards of measurement, physiological interpretation, and clinical use," *Circulation*, vol. 93, no. 5, pp. 1043–1065, 1996.
- [33] M. Brennan, M. Palaniswami, and P. Kamen, "Do existing measures of Poincaré plot geometry reflect nonlinear features of heart rate variability?," *IEEE Trans. Biomed. Eng.*, vol. 48, no. 11, pp. 1342–1347, 2001.
- [34] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *Proc. AAAI*, 2018.
- [35] M. Ibarra-Manzano *et al.*, "ITSC: Inter-turn short-circuit fault dataset for three-phase induction motors," GitHub repository, <https://github.com/ibarram/ITSC>.
- [36] R. Cunha, "MIT short-circuit flux and current signals dataset," Kaggle, 2021, <https://www.kaggle.com/datasets/rebecacunha/mit-short-circuit-flux-and-current-signals>.
- [37] A. Vaswani *et al.*, "Attention is all you need," in *Proc. NeurIPS*, 2017.
- [38] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv:1803.01271*, 2018.
- [39] A. Sharma *et al.*, "Recurrence-plot-based severity assessment of stator inter-turn faults using ensemble learning," *Measurement*, 2023.